



CMR UNIVERSITY

Private University Established in Karnataka State by Act No. 45 of 2013

SCHOOL OF ENGINEERING AND TECHNOLOGY

An Internship Report On “Data Management”

Submitted in partial fulfillment of the requirements for the award of degree in

Bachelor of Technology

in

Computer Science and Engineering

Of CMR University, (Lakeside Campus) Bangalore

Submitted by:

**Aatif Imaaz
22BBTCS003**

***Internship Carried out
at***

RIALZO LEATHERS LLP

*No.17, 1st Floor, 4th Cross, Masjid-E-Khaif Road, Bhimanna Layout, L.R. Bande, R.T. Nagar,
Nagar Post, Bengaluru, Karnataka - 560032.*

Internal Guide

Dr. Paramesaran.T

Professor,
Dept. of CSE, SOET

External Guide:

Mr. Aleem Ahmed

Manager,
RIALZO LEATHERS LLP

Department of Information Technology

**Off Hennur - Bagalur Main Road,
Near Kempegowda International Airport, Chalahatti,
Bangalore, Karnataka-562149**

2022-2023



CMR UNIVERSITY
Private University Established in Karnataka State by Act No. 45 of 2013

SCHOOL OF ENGINEERING AND TECHNOLOGY

Department of Information Technology

CERTIFICATE

This is to certify that the Internship work entitled “Data Management”, submitted to the CMR University, Bangalore, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a record of work done by Mr. Aatif Imaaz bearing university register number 22BBTCS003 during the academic year 2022–23 at School of Engineering and Technology, CMR University, Bangalore under my supervision and guidance. The Internship report has been approved as it satisfies the academic requirement in respect of internship work prescribed for the said degree.

Internal Guide:

Dr. Parameswaran T
Professor,
Dept. of CSE
SOET, Bangalore

External Guide:

Mr. Aleem Ahmed
Manager,
RIALZO LEATHERS LLP

Signature of the HOD
(Dr. P. Rubini)

Signature of the Dean
(Dr. N. Kannan)

Examiners Signature with date

(1) _____
Name

Signature

(2) _____
Name

Signature



Certificate

OF INTERNSHIP

PRESENTED WITH PROUD TO

Gopi Krishna

Has Successfully Completed Two months Of Training On
Cyber Security

From The Period Of

05-02-2023 to 30-03-2023

With Excellent Results, During His/Her Tenure With Us, We Found
Him/Her To Be A Keen And Enthusiastic Worker Who Made Valuable
Contributions To The Projects.



Certificate ID : CAMP 2178

ger

DECLARATION

I, **Aatif Imaaz** bearing USN **22BBTCS003**, student of Bachelor of Technology, Computer Science and Engineering, CMR University, Bengaluru, hereby declare that the Internship-I work entitled “Data Management” submitted by me, for the award of the Bachelor’s degree in Computer Science and Engineering to CMR University is a record of bonafide work carried out independently by me under the supervision and guidance of **Dr. Parameswaran T**, Professor, CSE Department, CMR University, Bangalore.

I further declare that the work reported in this internship work has not been submitted and will not be submitted, either in part or in full, for the award of any other degree in this university or any other institute or university.

Place: Bengaluru

Date:

AATIF IMAAZ
(USN:22BBTCS003)

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of this project would be incomplete without acknowledging the contributions of the people whose constant guidance, encouragement, and support made this work possible.

I consider myself privileged to express my sincere gratitude and respect to all those who guided me throughout the completion of this project. I express my heartfelt thanks to my Internal Internship Guide, **Dr. Parameswaran T**, Professor, Department of Computer Science and Engineering, School of Engineering and Technology, CMR University, for his continuous support, valuable guidance, and encouragement during the course of this internship.

I express my sincere gratitude to my External Internship Guide, **Mr Aleem Ahmed, manager, RIALZO LEATHERS LLP** without whose constant guidance and support the successful completion of this internship would not have been possible.

I would like to express my sincere thanks to **Dr. P. Rubini, Professor and Head**, Department of Computer Science and Engineering, School of Engineering and Technology, CMR University, Bangalore, for his encouragement and motivation throughout the internship period.

I express my heartfelt gratitude to **Dr. N. Kannan, Dean**, School of Engineering and Technology, CMR University, for his continuous support and encouragement.

I would also like to thank our Internship Coordinator, **Dr. Parameswaran T**, for his invaluable guidance, support, and encouragement throughout the tenure of this internship.

Aatif Imaaz

ABSTRACT

This internship report presents a detailed study of Data Management practices carried out at Rialzo Leathers LLP, with a focus on inventory management, data organization, and business decision support. The internship aimed to understand how structured data collection, cleaning, storage, and visualization can improve operational efficiency and support strategic decision-making. The work involved managing large volumes of inventory-related data, including raw material inputs, work-in-process, finished goods, wastage, and packaging materials. Tools such as Microsoft Excel were used for data entry, cleaning, validation, and automation, while Microsoft Power BI was utilized to create dashboards and visual reports for management analysis and real-time monitoring of inventory status.

The internship also emphasized the role of data analytics in improving resource allocation, reducing operational costs, and enhancing overall business performance. By transforming raw data into meaningful insights through reports and visualizations, the organization was able to track inventory levels accurately, identify inefficiencies, and support informed managerial decisions. This internship provided practical exposure to real-world data management challenges such as data quality, integration, and reporting, while strengthening analytical, technical, and problem-solving skills. Overall, the experience offered a strong foundation in data management concepts and demonstrated the importance of data-driven approaches in business

TABLE OF CONTENTS

TITLE	PAGE NO
DECLARATION	i
ACKNOWLEDGEMENT	ii
ABSTRACT.....	iii
TABLE OF CONTENTS.....	iv
LIST OF FIGURES	vi
CHAPTER	
1. INTERNSHIP OBJECTIVES.....	1
2. INTRODUCTION.....	3
2.1 ABOUT CAMPALIN.....	3
2.2 CYBER SECURITY	3
2.3 BASIC PENTESTING.....	4
2.4 BURP SUITE BASICS.....	4
2.5 HTTP IN DETAIL	4
2.6 INTRODUCTORY NETWORKING.....	4
2.7 KENOBI	5
2.8 LINUX FUNDAMENTALS.....	5
2.9 NMAP.....	5
2.10 OWASP JUICE SHOP	5
2.11 OWASP TOP 10	6
2.12 VULNVERSITY	6
3. INTERNSHIP DISCUSSION.....	7
3.1 INTRODUCTION TO ETHICAL HACKING AND CYBERSECURITY...7	
3.2 WHAT IS ETHICAL HACKING?	8
3.3 WHY IS ETHICAL HACKING IMPORTANT?	9
3.4 THE PHASES OF ETHICAL HACKING	10

3.5	TYPES OF ETHICAL HACKING	12
3.6	CYBERSECURITY THREATS AND VULNERABILITIES	13
3.7	IMPORTANT TECHNOLOGIES IN ETHICAL HACKING AND CYBER SECURITY.....	15
3.8	THE IMPORTANCE OF ETHICAL HACKING AND CYBERSECURITY	16
3.9	SKILLS LEARNT.....	17
3.10	WORK EXPERIENCE	18
3.11	CHALLENGES FACED	19
4.	CONCLUSION.....	20
5.	BIBLIOGRAPHY	21

LIST OF FIGURES

SL NO	FIGURES	PAGE NO
1	Fig. 3.1 Cybersecurity Process	7
2	Fig. 3.2 Ethical Hacking Threats	9
3	Fig. 3.3 Phases of Ethical Hacking	10
4	Fig. 3.4 Top Cybersecurity Threats	13

Chapter 1

INTERNSHIP OBJECTIVES

1. Data Collection and Organization:

- Understanding the fundamentals of data collection and management
- Learning how to gather data from multiple organizational sources
- Organizing raw data into structured formats for analysis

2. Microsoft Excel Basics:

- Understanding the role of Microsoft Excel in data management
- Learning to use formulas, functions, and data validation tools
- Practicing data entry, sorting, filtering, and basic automation

3. Data Cleaning and Preparation:

- Understanding the importance of data quality and accuracy
- Learning techniques to clean, remove duplicates, and correct errors
- Preparing datasets for analysis and reporting

4. Inventory Management System:

- Understanding inventory workflows and data flow within the organization
- Learning how to track raw materials, work-in-process, and finished goods
- Managing inventory levels to reduce wastage and operational costs

5. Data Analysis Techniques:

- Understanding basic data analysis concepts
- Learning to generate summaries and insights from collected data
- Analyzing trends to support business decision-making

6. Data Visualization Using Power BI:

- Understanding the role of data visualization in business analytics
- Learning to create dashboards and interactive reports
- Practicing visual representation of inventory and operational data

7. Report Generation and Documentation:

- Understanding how to convert data into meaningful reports
- Learning to present data according to management requirements
- Practicing preparation of quantitative and visual reports

8. Business Decision Support:

- Understanding how data supports strategic and operational decisions
- Learning to interpret reports for real-time decision-making
- Assisting management with data-driven insights

9. Data Security and Governance:

- Understanding the importance of data confidentiality and integrity
- Learning basic data governance and access control practices
- Ensuring safe handling of organizational data

10. Practical Industry Exposure:

- Gaining hands-on experience in real-world data management tasks
- Understanding organizational challenges related to large-scale data
- Developing analytical, technical, and problem-solving skills

CHAPTER 2

INTRODUCTION

2.1 ABOUT RIALZO LEATHERS LLP

Rialzo Leathers LLP is a manufacturing, processing, export, and import company specializing in premium lifestyle leather accessories for men and women. Incorporated on 28th March 2019, the organization is engaged in producing high-quality leather products for international brands while maintaining affordability and fashion relevance. The company focuses on quality assurance, operational efficiency, and customer satisfaction, and has established a strong presence in the global export market.

Rialzo Leathers LLP emphasizes structured operations, efficient inventory handling, and data-driven decision-making to manage its supply chain and production processes. The organization values integrity, transparency, and long-term relationships with clients and stakeholders, making effective data management a critical component of its business operations.

2.2 DATA MANAGEMENT

Data management refers to the process of collecting, organizing, storing, securing, and analyzing data to support business operations and decision-making. In modern organizations, large volumes of data are generated through production, inventory, sales, and logistics activities. Effective data management ensures data accuracy, consistency, and accessibility, enabling organizations to derive meaningful insights and improve operational efficiency.

At Rialzo Leathers LLP, data management plays a vital role in inventory control, resource allocation, cost reduction, and performance monitoring. Proper handling of data helps the organization track materials, reduce wastage, and support strategic planning through reliable reports and dashboard

2.3 DATA COLLECTION

During the internship, data was collected from various internal sources including inventory records, production logs, and management inputs. The focus was on gathering accurate data related to raw materials, work-in-process, finished goods, wastage, and packaging materials.

The collected data served as the foundation for analysis, reporting, and visualization, ensuring that management decisions were based on reliable and up-to-date information.

2.4 MICROSOFT EXCEL BASICS

Microsoft Excel was used extensively as a primary tool for data entry, organization, and analysis. During the internship, I gained practical experience in using Excel features such as formulas, sorting, filtering, data validation, drop-down lists, VLOOKUP functions, and basic automation.

Excel enabled efficient handling of large datasets and helped maintain structured inventory records, improving accuracy and ease of access to information.

2.5 DATA CLEANING AND PREPARATION

Data cleaning is an essential step in data management to ensure accuracy and reliability. During the internship, raw data was cleaned by removing duplicates, correcting inconsistencies, and validating entries to eliminate errors.

This process ensured that the data used for analysis and reporting was consistent, complete, and suitable for generating meaningful insights.

2.6 INVENTORY MANAGEMENT SYSTEM

The internship involved understanding and working with the organization's inventory management system. Inventory data included raw material inputs, vendor details, production stages, wastage, and finished goods ready for dispatch.

By organizing inventory data systematically, the system enabled real-time tracking of stock levels and supported efficient inventory planning and control.

2.7 DATA ANALYSIS

Basic data analysis techniques were applied to understand inventory trends, resource utilization, and operational performance. Summaries and comparisons were generated to identify inefficiencies and areas for improvement.

The analysis supported management in making informed decisions related to procurement, production planning, and cost optimization

2.8 DATA VISUALIZATION USING POWER BI

Microsoft Power BI was used to transform raw data into interactive dashboards and visual reports. These dashboards provided a clear overview of inventory levels, production status, and key performance indicators.

Data visualization helped management quickly interpret trends and patterns, enabling faster and more effective decision-making.

2.9 REPORT GENERATION

Based on the analyzed data, structured reports were generated to meet management and business requirements. These reports presented quantitative data in a clear and understandable format, supporting operational reviews and strategic discussions.

Accurate reporting ensured transparency and consistency in organizational decision-making.

2.10 BUSINESS DECISION SUPPORT

The internship emphasized the role of data in supporting business decisions. By converting raw data into meaningful insights, management was able to evaluate inventory utilization, reduce wastage, and improve operational efficiency.

Data-driven decision support enhanced planning accuracy and contributed to better resource allocation.

2.11 DATA SECURITY AND GOVERNANCE

Basic principles of data security and governance were followed to ensure confidentiality and integrity of organizational data. Access to sensitive data was handled responsibly, and data was maintained according to organizational guidelines. Proper data governance ensured reliability and compliance with internal policies.

2.12 PRACTICAL INDUSTRY EXPOSURE

The internship provided hands-on exposure to real-world data management challenges such as handling large datasets, ensuring data quality, and generating actionable insights. This experience helped in developing analytical, technical, and professional skills essential for a career in data management and analytics.

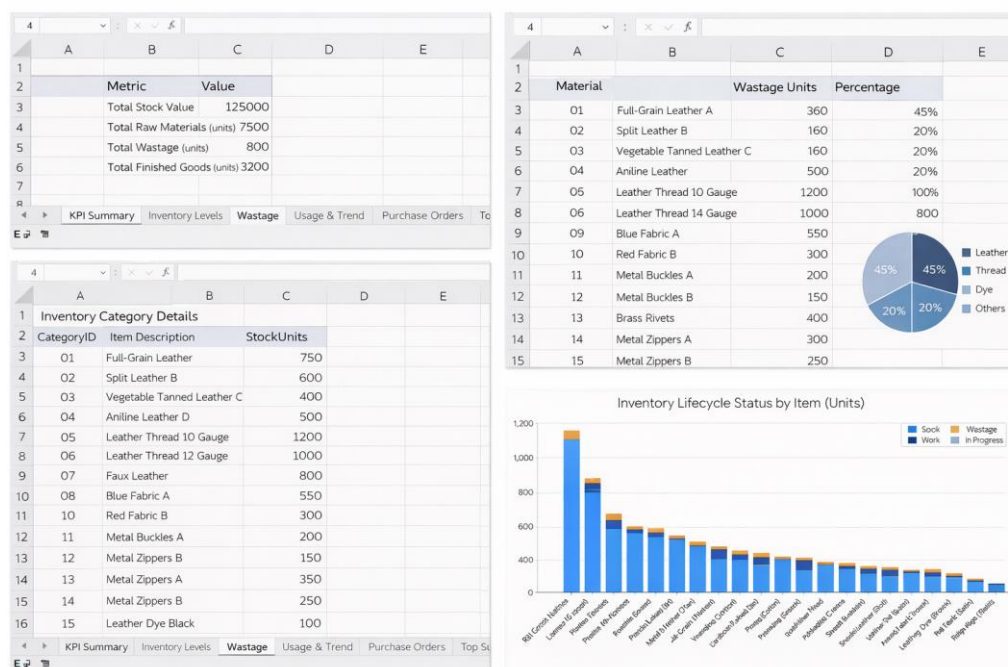
CHAPTER 3

INTERNSHIP DISCUSSION

3.1 INTRODUCTION TO DATA MANAGEMENT

In today's competitive and technology-driven business environment, data has become one of the most valuable assets for organizations. Every business activity—ranging from procurement and production to sales and logistics—generates large volumes of data on a daily basis. Managing this data effectively is essential for organizations to maintain efficiency, accuracy, and sustainability. Data management provides a structured approach to handling this information so that it can be transformed into meaningful insights that support business operations and decision-making.

In manufacturing organizations such as Rialzo Leathers LLP, data management plays a vital role in monitoring inventory levels, tracking production processes, minimizing wastage, and ensuring timely delivery of products. Without proper data management practices, organizations may face issues such as data inconsistency, duplication, operational delays, and poor decision-making. This internship focused on understanding and implementing data management techniques to support organizational goals and improve operational efficiency.



(Figure 3.1 INVENTORY DATA SUMMARY)

3.2 OVERVIEW OF DATA MANAGEMENT

Data management refers to the process of collecting, organizing, storing, securing, analyzing, and maintaining data throughout its lifecycle. The objective of data management is to ensure that data is accurate, consistent, accessible, and secure for authorized users. Effective data management allows organizations to derive insights, generate reports, and make informed decisions based on reliable information.

A well-structured data management system helps reduce manual errors, improves transparency, and enables better coordination between departments. In the context of this internship, data management was applied to inventory records, vendor details, production outputs, wastage tracking, and packaging material management. These activities highlighted the importance of maintaining clean and structured datasets to support operational and strategic planning

3.3 CONCEPT OF DATA MANAGEMENT

Data management is the process of collecting, organizing, storing, validating, securing, and analyzing data in a systematic manner. It involves multiple stages that ensure data quality and usability. Effective data management allows organizations to convert raw data into actionable insights that support business goals.

At its core, data management focuses on maintaining data integrity and reliability. This means ensuring that data is free from errors, duplication, and inconsistencies. In a manufacturing environment, even small data inaccuracies can lead to significant operational issues such as incorrect procurement quantities or inaccurate inventory levels. Therefore, structured data handling practices are essential.

During the internship, data management was applied to operational datasets including raw material inventory, production stages, finished goods, wastage records, and packaging materials. These datasets were continuously updated and required consistent validation to ensure accuracy. This experience demonstrated how proper data management serves as the backbone of efficient organizational operations.

3.4 ROLE OF DATA MANAGEMENT IN MANUFACTURING ORGANIZATIONS

Manufacturing organizations rely heavily on data to monitor and control their operations. Data management plays a vital role in ensuring that production schedules, inventory levels, and procurement activities are aligned with business objectives.

In Rialzo Leathers LLP, data management supported the following areas:

- Monitoring availability of raw materials
- Tracking work-in-process and finished goods
- Managing vendor supply timelines
- Identifying wastage and inefficiencies
- Supporting export and logistics planning

By maintaining structured datasets, the organization was able to reduce manual dependency and improve transparency across departments. This reinforced the importance of data as a strategic asset rather than just a record-keeping tool.

3.5 DATA MANAGEMENT LIFE CYCLE

The data management lifecycle consists of several interconnected stages that ensure data remains useful throughout its existence.

3.5.1 Data Creation and Collection

Data creation begins at the operational level where information is generated through daily activities. During the internship, data was collected from inventory registers, vendor records, production logs, and management inputs. Accurate data collection was emphasized to ensure reliability in subsequent stages.

3.5.2 Data Cleaning and Preprocessing

Raw data often contains inconsistencies, missing values, and duplicate entries. Data cleaning involved validating records, correcting errors, and standardizing formats. This step was crucial in improving data accuracy and usability.

3.5.3 Data Organization and Structuring

Cleaned data was organized into structured tables using spreadsheets. Proper structuring enabled efficient sorting, filtering, and retrieval of information. This stage ensured that data could be easily analyzed and reported.

3.6 INVENTORY DATA MANAGEMEN

NT

Inventory data management was a key focus area of the internship. Inventory data included information related to raw materials, semi-finished products, finished goods, and packaging materials. Managing this data accurately was essential to prevent shortages, overstocking, and wastage.

The internship involved tracking inventory movements, recording vendor inputs, and monitoring stock levels across different production stages. Structured inventory data helped management assess stock availability in real time and plan procurement accordingly.



(Figure 3.2 INVENTORY MANAGEMENT DASHBOARD)

3.7 VENDOR AND PROCUREMENT DATA

Vendor-related data plays a significant role in manufacturing operations. During the internship, vendor data such as supplier names, material quantities, delivery dates, and pricing details were maintained systematically.

Managing procurement data helped in evaluating supplier performance and ensuring timely availability of raw materials. This data also supported cost analysis and vendor comparison, contributing to better procurement decisions.

3.8 DATA QUALITY AND CONSISTENCY

Data quality refers to the accuracy, completeness, and reliability of data. Maintaining data quality was a continuous challenge during the internship due to frequent updates and multiple data sources.

To address this, validation checks were implemented, and standardized formats were followed. Ensuring data consistency helped in generating reliable reports and avoided discrepancies during analysis.

3.9 USE OF MICROSOFT EXCEL IN DATA MANAGEMENT

Microsoft Excel served as the primary tool for data handling during the internship. Excel was used for data entry, cleaning, sorting, filtering, and basic automation.

Advanced features such as formulas, VLOOKUP, drop-down lists, and conditional formatting were used to improve efficiency and accuracy. Excel enabled effective handling of large datasets and supported daily operational requirements.

Item Name	Received Amount	Final Goods	Work in Progress	Wastage	To be Processed
Full-Grain Leather (Tan)	800	450	200	80	70
Suede Leather (Blue)	500	280	110	60	50
Sheep Nappa (Black)	600	350	150	40	60
Lining Fabric (Satin)	400	250	80	20	50
Metal Buckles (Zinc)	1200	900	150	30	120
Premium YKK Zippers	1500	1100	200	50	150
Polyester Thread	500	380	60	25	35
Vegetable Tanned Hide	700	420	130	90	60
Cardboard Boxes (Large)	1000	850	0	40	110
Cardboard Boxes (Small)	2000	1800	0	70	130
Leather Dye (Black)	300	210	40	20	30
Leather Dye (Brown)	300	190	50	25	35
Gum/Adhesive	250	180	30	15	25
Sewing Machine Needles	100	85	0	5	10
Dust Bags (Cotton)	1200	1050	50	30	70
Buffing Wax	150	110	20	10	10
Edge Paint	200	140	30	15	15
Rivets (Brass)	3000	2600	200	60	140
Magnetic Clasps	400	320	40	15	25
Polishing Cloth	500	460	0	10	30
Packaging Tape	600	550	0	15	35
Silica Gel Packets	5000	4800	0	40	160
Branding Labels (N/H)	800	720	30	10	40
Foam Inserts	1000	900	40	20	40
Shoulder Strap Chains	300	240	30	10	20
TOTAL	24300	19235	1640	715	1560

(Figure 3.3 DETAILED INVENTORY STATUS)

3.10 DATA CLEANING AND PREPARATION TECHNIQUES

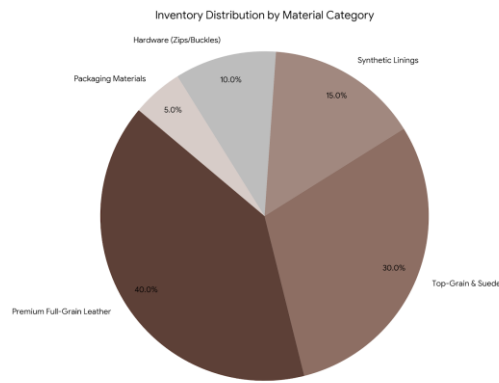
Data cleaning is one of the most crucial stages in the data management process, as the accuracy of analysis depends entirely on the quality of data being used. During the internship, raw data obtained from different operational sources often contained inconsistencies, missing values, and duplicate entries. These issues required careful examination and correction before the data could be used for meaningful analysis.

The data cleaning process involved validating inventory quantities, standardizing vendor names, correcting date formats, and ensuring consistency in units of measurement. Missing values were identified and resolved through cross-verification with source records or management inputs. Duplicate records were removed to prevent inflated values and reporting errors. Through these activities, the reliability of the datasets was significantly improved, enabling accurate reporting and analysis.

3.11 DATA ORGANIZATION AND STRUCTURING

Once cleaned, data was organized into well-defined structures to facilitate ease of access and analysis. Data organization involved categorizing information into logical groups such as raw material inventory, work-in-process, finished goods, and packaging materials. Each category was maintained in separate but interconnected datasets to allow clear visibility across different operational stages.

Structured data organization enabled effective sorting, filtering, and summarization. This approach reduced complexity and improved clarity when handling large datasets. Well-organized data also supported faster retrieval of information, saving time and effort during reporting and decision-making processes.



(Figure 3.4: Inventory Distribution by Material Category)

3.12 DATA QUALITY ISSUES AND OPERATIONAL CHALLENGES

During the internship, several data quality issues were encountered while working with organizational datasets. Data quality plays a crucial role in data management, as inaccurate or inconsistent data can significantly affect analysis results and business decisions. In a manufacturing organization like Rialzo Leathers LLP, data is generated from multiple operational points such as inventory intake, production stages, and dispatch records, which increases the likelihood of errors.

One of the major challenges observed was inconsistency in data entry. Similar items were sometimes recorded using different names or formats, which made aggregation and comparison difficult. Missing values were also found in certain inventory and procurement records, requiring additional verification with physical documents or management inputs. Duplicate entries in some datasets caused discrepancies in inventory totals and reporting outcomes.

To address these issues, data validation techniques were applied, including standardizing formats, removing duplicate records, and cross-checking values with source data. These challenges emphasized the importance of structured data entry guidelines, regular audits, and disciplined data handling practices. Resolving data quality issues improved the accuracy and reliability of reports generated during the internship.

1	Inventory Category Details			
2	CategoryID	ItemDescription	StockUnits	
3	01	Full-Grain Leather A	750	
4	02	Split Leather B	600	
5	03	Vegetable Tanned Leather C	400	
6	04	Aniline Leather D	500	
7	05	Faux Leather E	300	
8	06	Leather Thread 10 Gauge	1200	
9	07	Leather Thread 12 Gauge	1000	
10	08	Leather Thread 14 Gauge	800	
11	09	Blue Fabric A	550	
12	10	Red Fabric B	300	
13	11	Metal Buckles A	200	
14	12	Metal Buckles B	150	
15	13	Brass Rivets	400	
16	14	Metal Zippers A	300	
17	15	Metal Zippers B	250	
18	16	Leather Dye Black	100	
19	17	Leather Dye Brown	80	
20	18	Conditioning Wax	120	
21	19	Shipping Boxes (Large)	200	
22	20	Shipping Boxes (Medium)	180	
23	21	Shipping Boxes (Small)	120	
24	22	Poly Bags (Large)	350	

(Figure 3.5: Inventory Category Details and Stock Units)

3.13 DATA INTEGRATION AND INTER-DEPARTMENTAL COORDINATION

Data integration refers to the process of combining data from multiple sources to create a unified view of organizational operations. During the internship, it was observed that different departments maintained data independently, which sometimes resulted in fragmented and disconnected datasets. Inventory, production, and procurement data were often stored separately, making it difficult to analyze operations holistically.

Integrating data across departments required aligning data formats, matching identifiers, and ensuring consistency between datasets. For example, linking vendor data with inventory records helped evaluate supplier performance and material inflow patterns. Similarly, integrating production data with inventory levels provided better visibility into material consumption and wastage.

Effective inter-departmental coordination was essential for successful data integration. Regular communication and verification helped reduce discrepancies and improve data consistency. This experience highlighted the importance of centralized data management systems and collaboration across departments to support efficient decision-making.



(Figure 3.6: Materials Usage and Wastage Analysis Dashboard)

3.14 DATA ANALYSIS FOR OPERATIONAL INSIGHTS

Data analysis was carried out to extract meaningful insights from the organized datasets. The analysis focused on understanding inventory trends, material utilization, and production efficiency. By examining historical and current data, patterns related to stock movement and consumption rates were identified.

Inventory analysis helped determine optimal stock levels and highlighted instances of overstocking and shortages. Production data analysis revealed areas where wastage was higher than expected, allowing management to investigate possible causes such as process inefficiencies or handling issues. These insights supported informed decisions related to procurement planning, inventory control, and cost reduction.

The analytical process demonstrated how structured data analysis can transform raw data into valuable information that supports operational improvements and strategic planning.

3.15 ROLE OF DATA MANAGEMENT IN BUSINESS DECISION-MAKING

Data management plays a vital role in supporting business decision-making by providing accurate, timely, and relevant information. During the internship, it became evident that data-driven decisions were more effective than decisions based solely on intuition or manual observation.

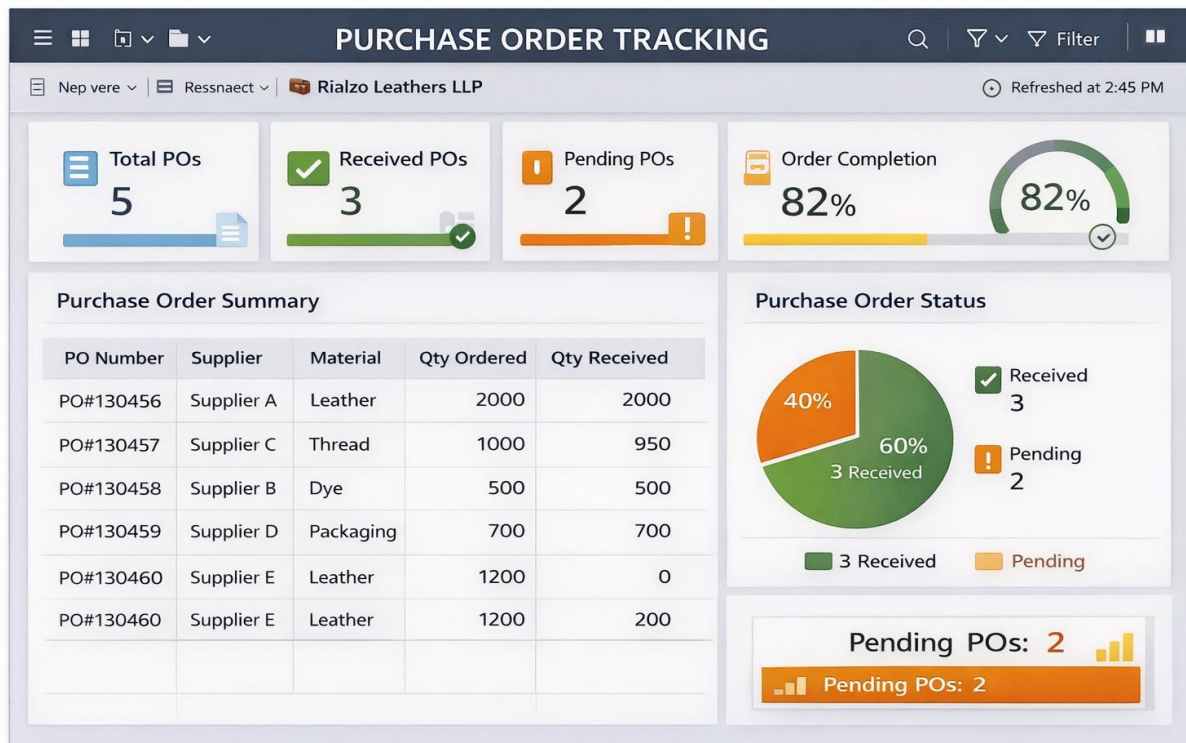
By maintaining organized datasets and generating analytical reports, management was able to assess inventory performance, monitor production efficiency, and allocate resources more effectively. Data visualization tools further enhanced decision-making by presenting insights in a clear and understandable format.

This experience highlighted how effective data management improves transparency, reduces uncertainty, and enables organizations to respond quickly to operational challenges and market demands.

3.16 OVERALL LEARNING OUTCOMES AND INTERNSHIP EXPERIENCE

The internship provided valuable practical exposure to real-world data management practices in a manufacturing environment. Working with actual organizational data helped bridge the gap between academic knowledge and industry application. The experience enhanced understanding of data collection, cleaning, integration, analysis, and reporting processes.

The internship also contributed to the development of technical skills in tools such as Microsoft Excel and Microsoft Power BI, as well as professional skills such as problem-solving, time management, and analytical thinking. Overall, the internship strengthened confidence and provided a solid foundation in data management concepts, preparing for future roles in data analytics and information system



(Figure 3.6: Purchase Order Tracking and Status Dashboard)

CHAPTER 4

CONCLUSION

The internship provided a comprehensive understanding of data management practices and their significance in a real-world manufacturing and export-oriented organization. Through hands-on exposure at Rialzo Leathers LLP, the internship successfully bridged the gap between theoretical knowledge gained in academics and practical application in an industry environment. The experience highlighted how structured data handling plays a crucial role in improving operational efficiency, accuracy, and decision-making.

One of the key outcomes of the internship was gaining practical knowledge of the complete data management lifecycle, including data collection, cleaning, organization, analysis, visualization, and reporting. Working with inventory, vendor, and production-related data emphasized the importance of maintaining data accuracy and consistency. Even minor discrepancies in data were found to have a significant impact on reporting outcomes, reinforcing the need for disciplined data handling and validation practices.

The use of tools such as Microsoft Excel and Microsoft Power BI enhanced technical proficiency and demonstrated how technology can simplify complex data processes. Excel automation helped reduce manual effort and improve efficiency, while Power BI dashboards enabled effective visualization of key metrics such as inventory levels, production status, and wastage. These tools played a vital role in transforming raw data into meaningful insights that supported managerial decision-making.

Overall, the internship was a valuable learning experience that strengthened foundational knowledge in data management and analytics. It provided practical exposure to real organizational challenges and prepared for future roles in data-related domains. The skills and insights gained during this internship will serve as a strong base for further academic learning and professional growth in the field of data management and information systems.

CHAPTER 5

BIBLIOGRAPHY

- [1] Han, J., Kamber, M., & Pei, J. (2012). *Data Mining: Concepts and Techniques*. Morgan Kaufmann Publishers, USA.
- [2] Silberschatz, A., Korth, H. F., & Sudarshan, S. (2019). *Database System Concepts*. McGraw-Hill Education, New York.
- [3] Kimball, R., & Ross, M. (2013). *The Data Warehouse Toolkit: The Definitive Guide to Dimensional Modeling*. Wiley Publishing.
- [4] Microsoft Corporation. (2023). *Microsoft Excel Documentation*.
<https://learn.microsoft.com/en-us/excel>
- [5] Microsoft Corporation. (2023). *Power BI Documentation*.
<https://learn.microsoft.com/en-us/power-bi>
- [6] Laudon, K. C., & Laudon, J. P. (2020). *Management Information Systems: Managing the Digital Firm*. Pearson Education.
- [7] Chopra, S., & Meindl, P. (2016). *Supply Chain Management: Strategy, Planning, and Operation*. Pearson Education.
- [8] Davenport, T. H., & Harris, J. G. (2017). *Competing on Analytics: The New Science of Winning*. Harvard Business Review Press.
- [9] Oracle Corporation. (2022). *Data Management Best Practices*.
<https://www.oracle.com/database/what-is-data-management/>
- [10] IBM Corporation. (2022). *Introduction to Data Management and Analytics*.
<https://www.ibm.com/topics/data-management>

Internship



Date :	
Week # :	1

Student ID	22BBTCS1
------------	----------

Internship Title	DATA MANAGEMENT
------------------	-----------------

Work completed this week	Data cleaning and validation
	Removing duplicate and inconsistent records
	Learning Excel formulas and data validation
	Understanding inventory flow and production stages
Deliverables submitted this week	Cleaned and validated inventory datasets
	Updated Excel sheets with formulas
contribution for the week	Identified data inconsistencies and corrected errors
	Improved data accuracy and reliability
Plan for next week	Learning data analysis and visualization tools
Progress since last report	-
Comment by Guide	
Signature of Students with Date	
Signature of Guide with Date	

Internship

Date :	
Week # :	2

Student ID	22BBTCS355
------------	------------

Internship Title	DATA MANAGEMENT
------------------	-----------------

Work completed this week	Learning Microsoft Power BI basics
Deliverables submitted this week	Basic Power BI dashboard
contribution for the week	Understood data visualization and reporting concepts
Plan for next week	Advanced dashboard creation and reporting
Progress since last report	Gaining confidence in data analysis and visualization tools
Comment by Guide	
Signature of Students with Date	
Signature of Guide with Date	

Internship

Date :	
Week # :	3

Student ID	22BBTCS335
------------	------------

Internship Title	DATA MANAGEMENT
------------------	-----------------

Work completed this week	Preparing reports for management review
	Enhancing Power BI dashboards with filters and visuals
Deliverables submitted this week	Detailed inventory and wastage analysis report
contribution for the week	Provided data-driven insights to support operational improvements
Plan for next week	Finalizing reports and documentation
Progress since last report	Improved analytical thinking and dashboard design skills
Comment by Guide	
Signature of Students with Date	
Signature of Guide with Date	